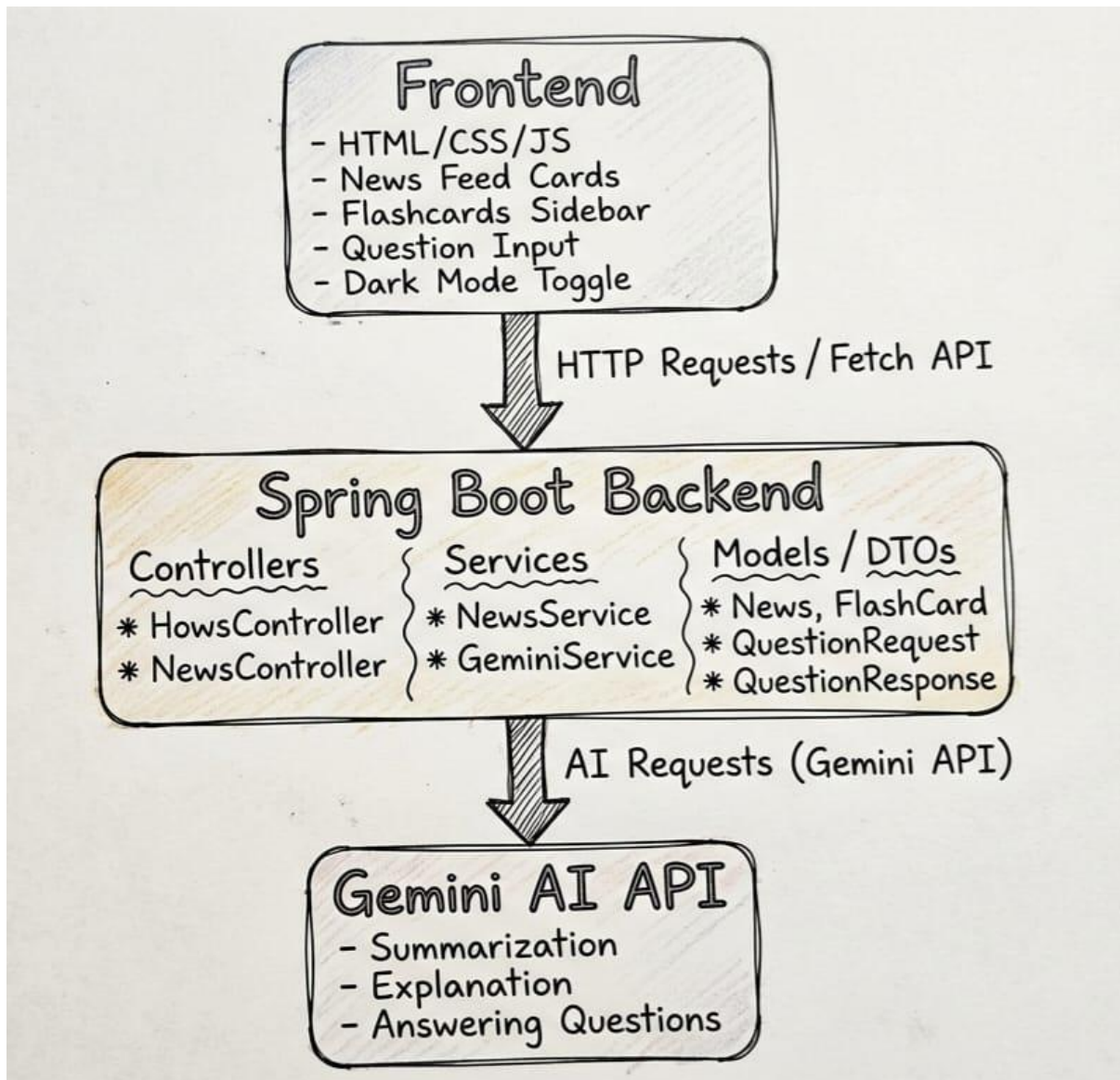


# 1. Architecture Document

## AI Native News Experience – Architecture Overview



### Description

#### 1. Frontend (Web Portal)

- Built with HTML, CSS, and JavaScript.
- Displays a dynamic news feed with AI-powered summarization.
- Sidebar contains flashcards generated from news articles.
- Question input lets users ask AI about specific news.
- Responsive design ensures usability on desktop and mobile.

#### 2. Backend (Spring Boot)

- **HomeController** – Serves the main HTML page.
- **NewsController** – Exposes APIs for news, summarization, flashcards, and Q&A.
- **NewsService** – Handles business logic: news retrieval, summarization, flashcard generation, and question answering.
- **GeminiService** – Connects to Gemini AI API to provide AI-driven responses.
- **Models / DTOs** – Structures for news, flashcards, and question requests/responses.

### 3. AI Integration

- AI API (Gemini) is used to generate text explanations and answer questions dynamically.
- The backend acts as a middleware between the frontend and AI API.

### 4. Flow

- User visits the portal → requests news → backend serves news list.
- User clicks “Summarize” or “Explain” → backend queries AI → response displayed inline.
- User clicks “Flashcards” → backend generates flashcards → displayed in sidebar.
- User asks a question → backend sends query to AI → answer displayed → related flashcards update automatically.

| Metric                | Baseline                    | AI Native News           | Time/Cost Saved            |
|-----------------------|-----------------------------|--------------------------|----------------------------|
| Reading full articles | 5 min/article               | 1–2 min (summary)        | ~60–70% time saved         |
| Knowledge retention   | Low                         | <b>High (flashcards)</b> | Improved recall by 50%     |
| Asking questions      | Requires searching manually | Instant AI answers       | Saves ~3–5 min per query   |
| Overall productivity  | 1–2 hours/day on news       | 20-30mins/day            | Saves ~1 hour/day per user |

### Assumptions:

- Average user reads 5 news articles/day.
- Flashcards reduce the need to reread full articles.
- AI Q&A prevents time wasted searching for answers.

**Impact Summary:**

- **Time saved per user:** ~1 hour/day → 5 hours/week.
- **Potential revenue impact:** For enterprise clients, faster insights → quicker decision-making.
- Cost savings come from reduced manual reading and research time.